Tooling Support to join Mojaloop Schemes

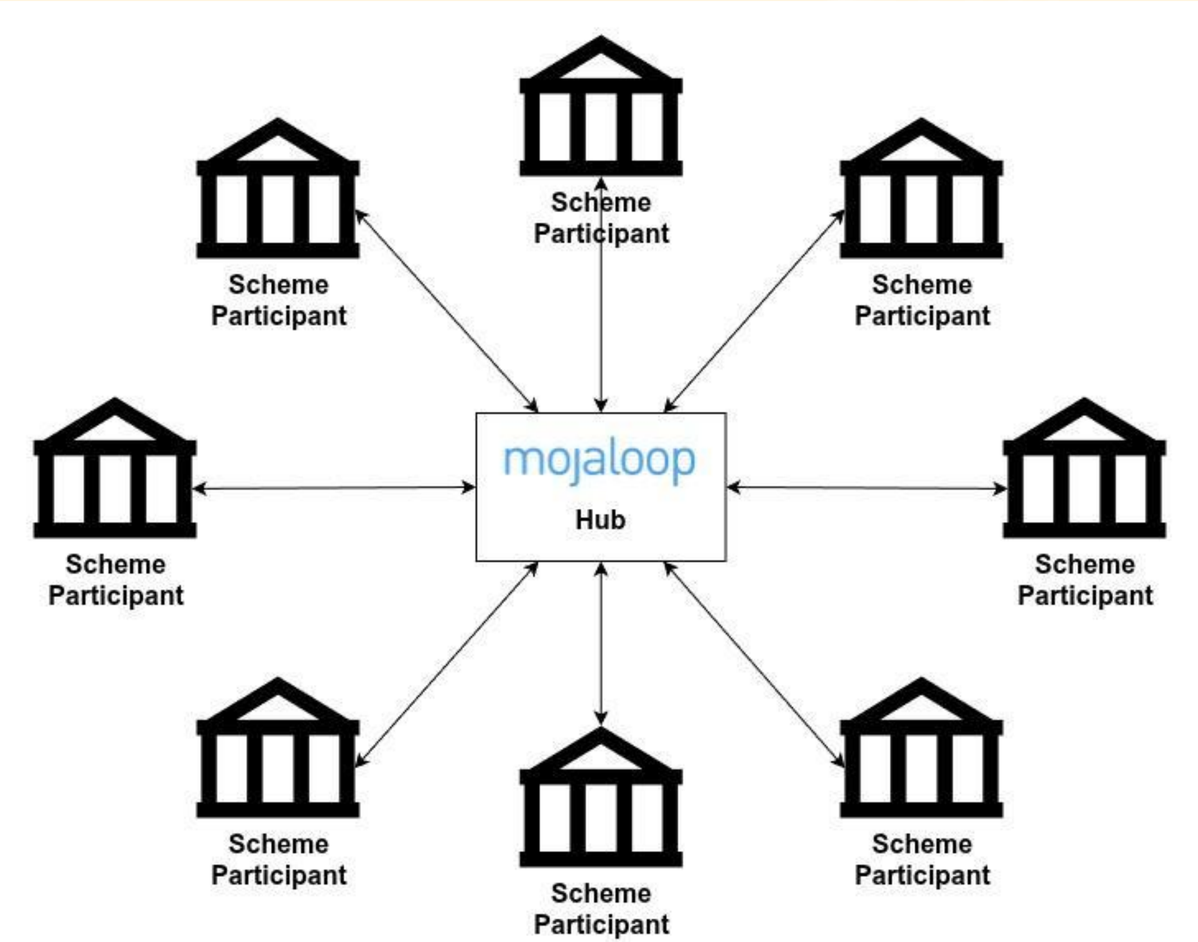
Open-source tools for hub operators and scheme participants



Some background...

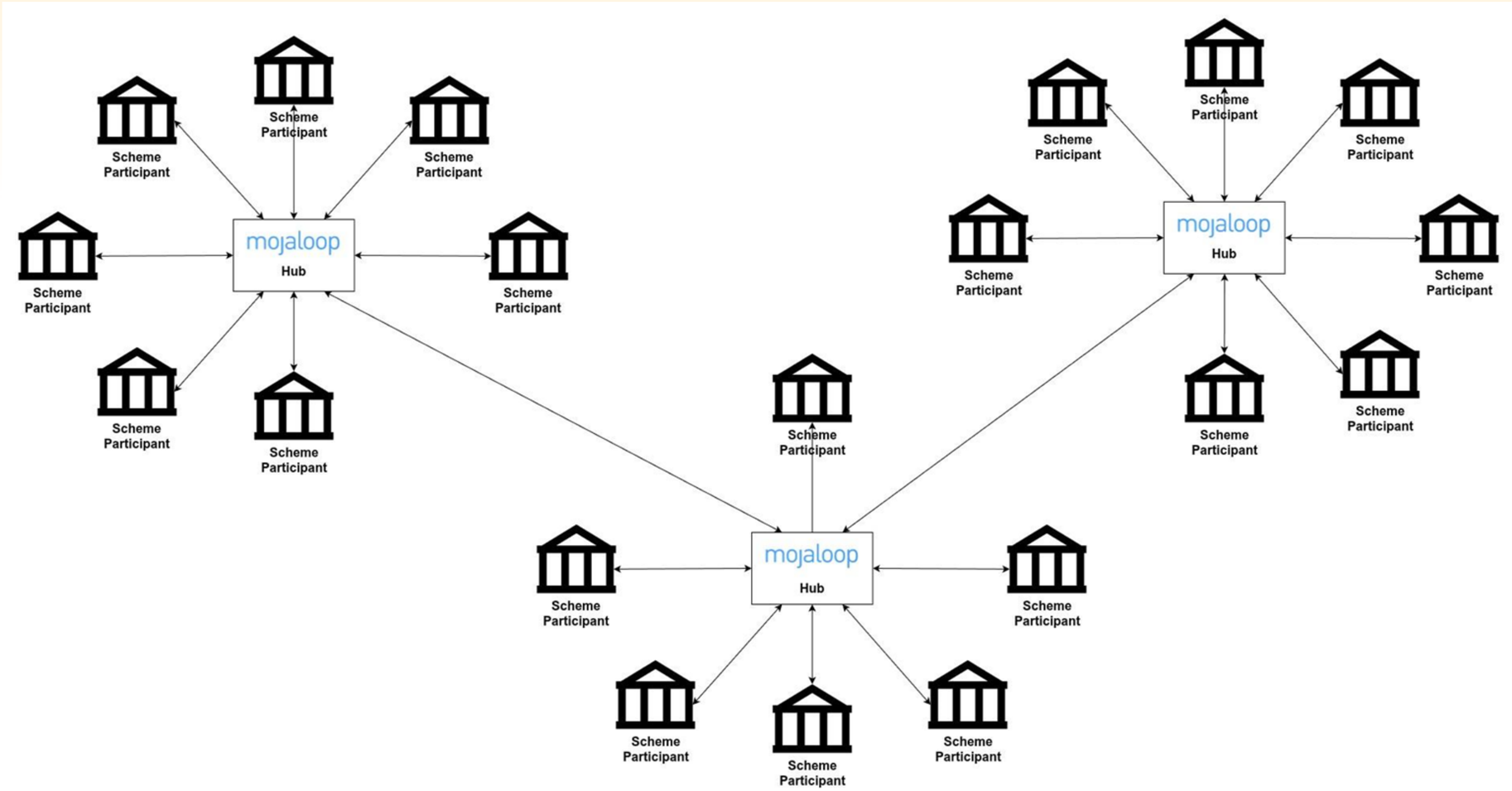
Typical Integration Architecture

Mojaloop schemes follow a “hub and spoke” architecture.

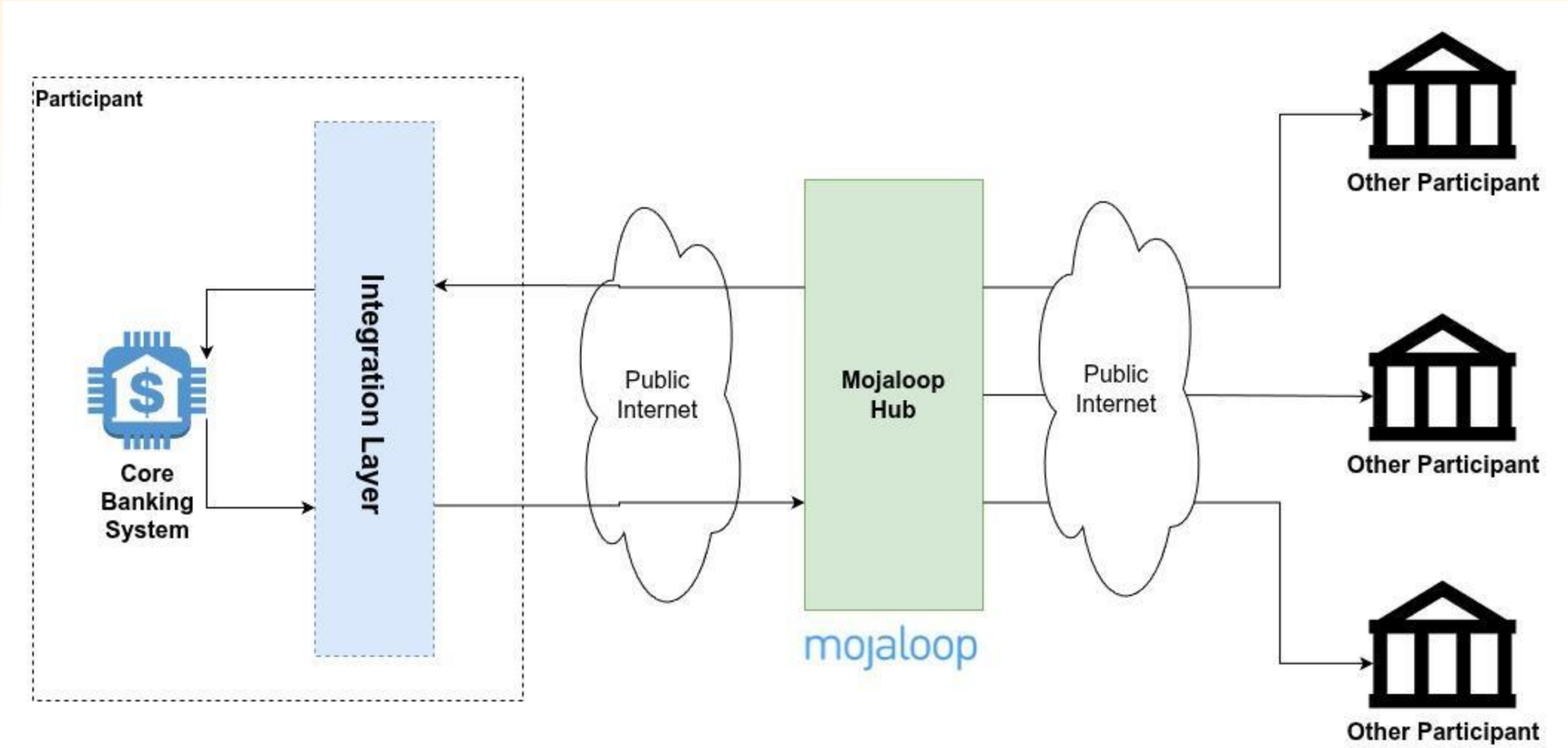


Typical Integration Architecture

Schemes can be interconnected...



Typical Integration Architecture



Banks, MoMos, Fintechs etc… typically connect to Mojaloop schemes by hosting an “integration layer” within their infrastructure to connect their “backend” to instance payment scheme APIs

Open-Source Integration Tools

- **Developing custom integration layers can be time consuming, expensive and error prone.**

- *Example: ~6 months elapsed time for a 3 person, dedicated participant software engineering team; plus ~3 months debugging*

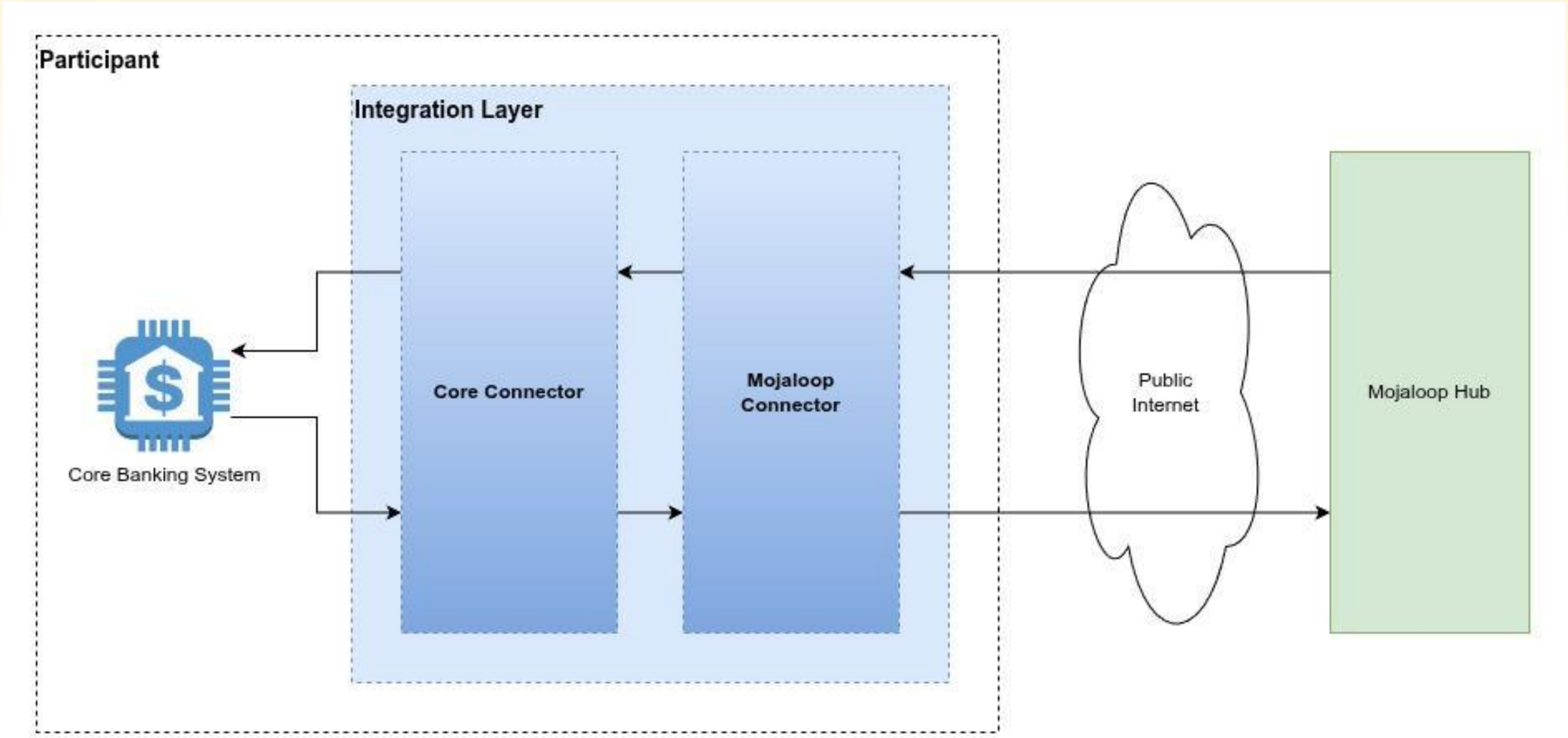
- To speed up, simplify and de-risk the participant integration process, several Mojaloop specific, open-source tools exist in the ecosystem

- *Example: ~2 weeks elapsed time for a single engineer*

Open-Source Integration Tools

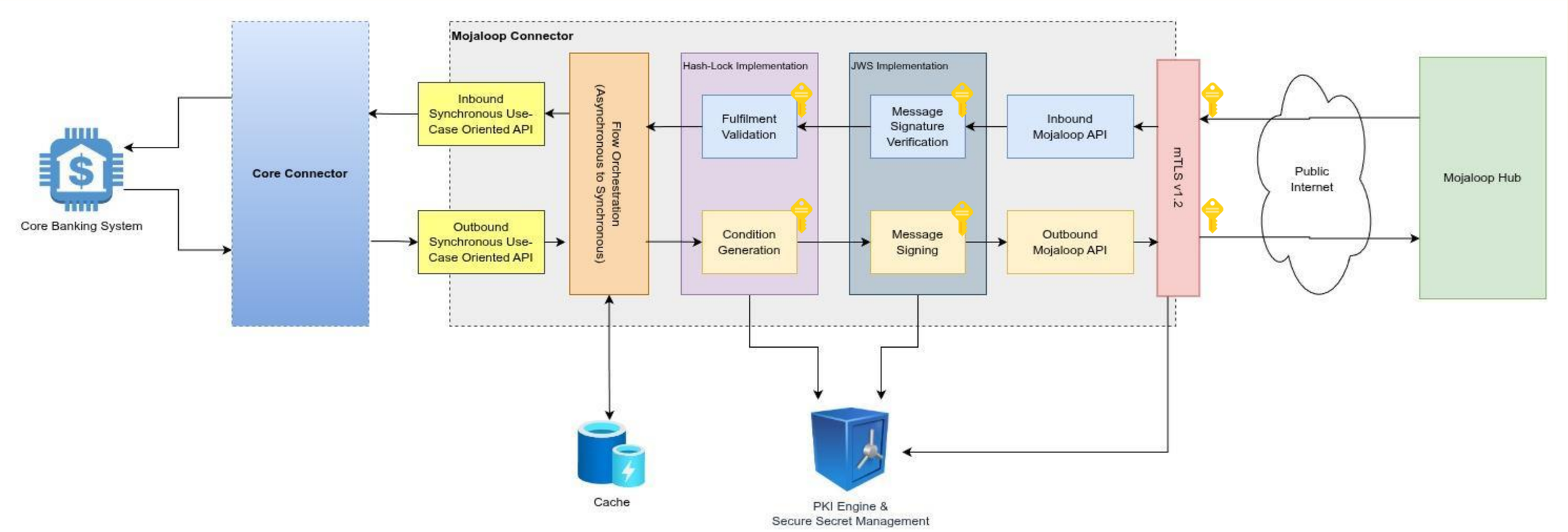
Tool Name	Description	Suitable for
Mojaloop Connection Manager (MCM)	*A hub and client side orchestration engine for managing the secure connections between participant systems and a Mojaloop hub*	*All Mojaloop adopters and participants.*
Mojaloop Connector (SDK)	A participant side abstraction service that simplifies API level integration between core banking systems and Mojaloop hubs	Participants with internal software engineering capability who want low level integration with Mojaloop APIs
Integration Toolkit (ITK)	A light-weight integration mechanism targeted at smaller participants with resource constraints	SME type participants who want a standardised integration with minimum effort, risk and infrastructure cost
Payment Manager for Mojaloop (PM4ML)	A feature rich enterprise level integration solution targeted at larger participants	Larger participants with more demanding requirements
Mifos Payment Hub EE	A multi-tenant integration solution aimed at aggregators	Aggregators who want to offer managed integrations to Mojaloop schemes

Typical Integration Architecture



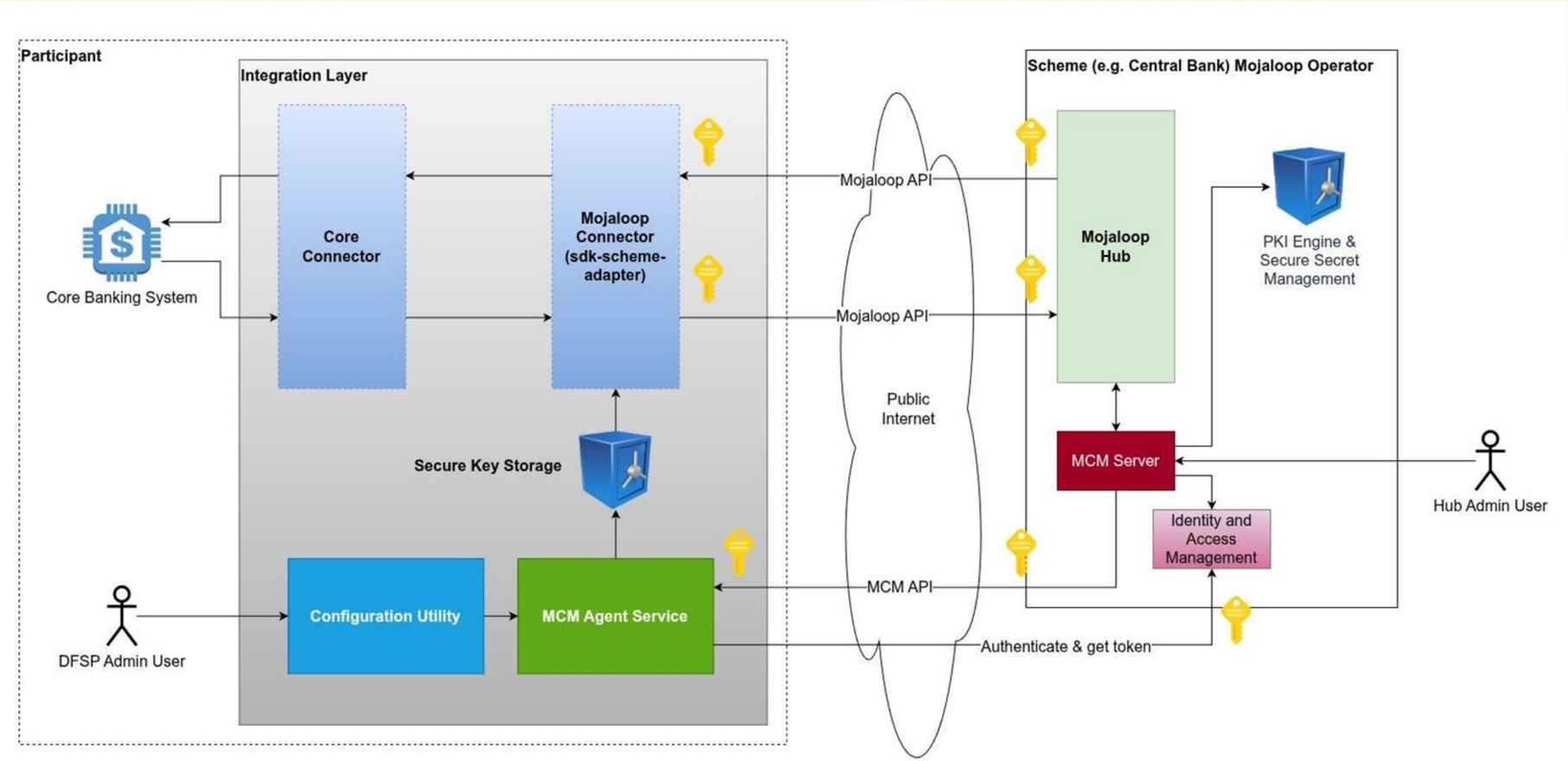
Mojaloop ITK and PM4ML follow a “2 connectors” pattern. One abstracts the Mojaloop API and connection security; the other abstracts the core banking system API.

Typical Integration Architecture



The Mojaloop connector abstracts and simplifies the generic transaction flows and asynchronous, resource-oriented API details by exposing a use-case oriented API e.g. “*send money*”, “*disburse loan*”, “*loan repayment received*”. The complexities of connection security, cryptography, message signing, signature verification and flow orchestration are managed.

Typical Integration Architecture



Mojaloop Connection Manager works with the “2 connectors” pattern to automate the process of secure connection setup and maintenance e.g. Secure keypair generation, PKI certificate generation, certificate signing and cross-signing exchanges.

Typical Integration Architecture

By using off-the-shelf, tried and tested abstractions we reduce the technical and business risks, effort, time and cost required for participants of all sizes to participate on equal terms in Mojaloop based payment schemes.

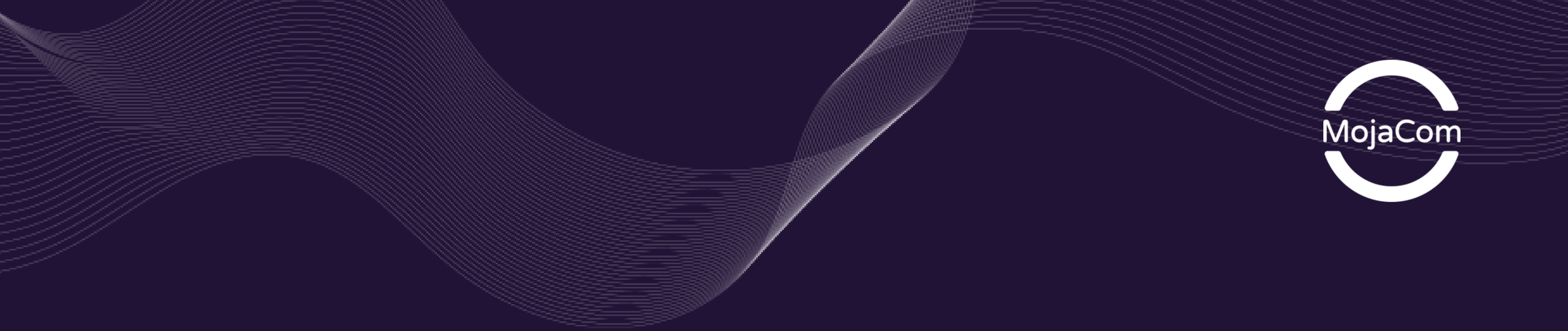
Participation Tools Workstream

- The participation tools workstream is where the Mojaloop community works on this set of open-source integration tools
  - Mojaloop Slack: **#ws-participation-tools**
  - Weekly meeting (zoom) on **Wednesdays @ 0800 UTC**

- Current Workstream focus:
  - Full adoption of donated PM4ML code base (thank to Infitx!)
  - More flexible integration of MCM into deployment tools (IaC)
  - Integrate donated System Integrator documentation (thank to Infitx!)

Progress this PI

- Adoption of PM4ML code base is approx 80% complete – some CI/CD remains
- Successful integration of MCM into IaC-Lite.
  - Refactoring of RBAC backend from Keycloak to full Ory stack
  - Alignment with other Mojaloop web portal architecture.
- Donated System Integrator documentation remains to be done
- A new integration tool has been donated by AbliPay…



Thank you for listening!

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